Scalability & Future Plan

Date	8 July 2026
Team ID	
Project Name	AI Student Evaluation Dashboard
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Local thread-bound scoring evaluation context cycles.	Concurrent evaluation queries experience queuing delay jumps under continuous multi-user traffic.	High
2	In-memory UI session state caching without non-volatile storage backing.	Refreshing user web browser viewports resets active analytical history metrics and active evaluation logs.	Medium
3	Monolithic rules engine definitions parsed locally.	Adding or scaling technical grading criteria requires modifying core application source files.	Low

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Tethered to single-instance synchronous Streamlit thread runtimes.	Deploy behind an Nginx reverse proxy with load balancing across autoscaling container tasks.
2	Data Storage	Volatile localized runtime variables and state context.	Integrate an external transactional persistent layer (PostgreSQL) paired with Redis for low-latency session caching.
3	Performance	Synchronous compute thread execution steps for analytical evaluations.	Implement asynchronous task queues via Celery and RabbitMQ to process intensive text verification background loops.
4	Security	Bypassed open sandbox environment without login credentials blocks.	Introduce structured JWT-based role authentication gates and encrypt text data transmissions over HTTPS.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Persistent historical analysis recording tracking models.	1 Month	Allows students and instructors to view historical score progression charts over time.
Phase 3	Distributed microservices decomposition of evaluation modules.	3 Months	Isolates UI rendering traffic entirely from backend deep-parsing NLP computations.
Phase 4	Multi-tenant role architecture for institutional scale deployment.	6 Months	Supports separate workspace administration for multiple educational branches and academic faculties.